THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 9: **The Amazing World of Solutes, Solvents and Solutions** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 20 per cent	40/200 x 100.
2	**(d)** colloid	Tyndall effect.
3	**(d)** more concentrated	More solute.
4	**(a)** 30 g	0.12 x 250.
5	**(b)** milk	Colloid.
6	**(c)** temperature and amount of sugar	Constants.
7	**(b)** 475 g	Salt = 25 g.
8	**(a)** sand in water	Suspension.
9	**(c)** 4 g remains undissolved	Saturation reached.
10	**(a)** 10 per cent	15/150 x 100.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Solution = 100 g (1); Percentage = 18 per cent (1).	1+1
12	Stirring only speeds up dissolving (1); Solubility depends on temperature and nature (1).	1+1
13	Lemon water is a true solution with tiny particles (1); Milk particles are larger and scatter (1).	1+1
14	8/100 x 250 (1); = 20 g (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Mass of solution = 175 g (1); 35/175 x 100 (1); = 20 per cent (1).	1+1+1
16	Prepare a hot saturated solution (1); Filter and cool slowly (1); Crystals form and are dried (1).	1+1+1
17	Evaporation (1); Distillation (1); Crystallisation (1).	1+1+1
	OR: Water is 90 per cent (1); Solution = 360/0.9 = 400 g (1); Sugar = 40 g (1).	1+1+1
18	Oil is less dense and unlike water (1); Does not mix with water (1); Dissolves in kerosene (1).	1+1+1
	OR: It has no dissolved salts (1); Avoids deposits and scale (1); Improves life of equipment (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Variable: temperature of water (1); Same volume of water in each beaker (1); Add solute until saturated and record the mass (1); Table of masses at 20, 40, 60 degrees Celsius (1); Both rise with temperature, sugar more (1).	1+1+1+1+1
	OR: Solubility decreases (1); Crystals separate (1); Remaining solution stays saturated (1); Mass of crystals depends on the difference (1); Used in crystal preparation (1).	1+1+1+1+1
20	Original = 16 per cent (1); After evaporating: 40/200 = 20 per cent (1); After adding water: 40/350 = 11.4 per cent (1); Evaporation raises concentration (1); Dilution lowers it (1).	1+1+1+1+1
	OR: Lab: distillation (1); Heating and condensing (1); Nature: evaporation from sea (1); Condensation into clouds (1); Rain (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 9: **The Amazing World of Solutes, Solvents and Solutions** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** a suspension	Large particles.
2	**(a)** unsaturated	Higher solubility.
3	**(b)** 15 g	0.25 x 60.
4	**(b)** purple and homogeneous	Solution.
5	**(b)** cold water under pressure	Gas solubility.
6	**(a)** 50 g	10 / 0.2.
7	**(c)** comes out as crystals	Less solubility.
8	**(c)** all of it	To dryness.
9	**(a)** decreases	Larger total mass.
10	**(d)** solubility increases with temperature	Higher solubility.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Dissolved gas escapes as pressure falls (1); More gas dissolves at high pressure (1).	1+1
12	Taste or evaporate (1); Residue of sugar remains (1).	1+1
13	They are of molecular size (1); Much smaller than the limit of visibility (1).	1+1
14	Dissolving needs a solvent (1); Melting is a change of state by heat (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Salt = 30 g (1); Water = 170 g (1); Dissolve salt in water (1).	1+1+1
16	Powder has more surface area (1); Faster dissolution (1); Same quantity of sugar dissolves finally (1).	1+1+1
17	Sugar forms a solution; sand a suspension (1); Sugar is not visible; sand is (1); Sand settles; sugar does not (1).	1+1+1
	OR: Solute = 10 g (1); New solution = 100 g, so 10 per cent (1); The solution is more dilute (1).	1+1+1
18	C is most concentrated (1); A solution = 105 g (1); A: 5/105 x 100 = 4.76 per cent (1).	1+1+1
	OR: Solubility rises sharply with temperature for KNO3 (1); Cooling a hot saturated solution gives crystals (1); Used in crystal preparation (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Size: <1 nm, 1-1000 nm, >1000 nm (1); Filtration (1); Settling (1); Tyndall effect (1); Examples (1).	1+1+1+1+1
	OR: Particles fit in gaps (1); Mass is conserved (1); Example: alcohol and water (1); Salt in water (1); Shows gaps in matter (1).	1+1+1+1+1
20	Water dissolves nutrients (1); Carries wastes (1); Dissolves oxygen for fish (1); Warm and polluted water holds less oxygen (1); Fish die (1).	1+1+1+1+1
	OR: Pressure falls on opening (1); Dissolved CO2 comes out as bubbles (1); Hiss from rapid escape (1); Gas lost from the drink (1); Warm drink loses it faster (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 9: **The Amazing World of Solutes, Solvents and Solutions** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** a separating funnel	Immiscible.
2	**(a)** 5 per cent	10 g / 200 g.
3	**(a)** dissolved air coming out	Gas solubility falls.
4	**(b)** alcohol or water	Solvent.
5	**(d)** 13.3 per cent	20 g / 150 g.
6	**(b)** evaporation	Dissolved solute.
7	**(d)** surface area of the solid	More surface.
8	**(b)** 200 g per 100 g of water	Definition.
9	**(d)** acetone	Like dissolves like.
10	**(a)** solubility increases with temperature	Temperature effect.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Gas solubility decreases with temperature (1); Cold water retains more gas (1).	1+1
12	Hot water particles move faster (1); Solubility is higher (1).	1+1
13	Total 100 g (1); 4 per cent (1).	1+1
14	Acid with less solute (1); Add acid slowly to water (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	True solution: tiny particles transmit light (1); Suspension: large particles block or scatter (1); Particles in suspension settle (1).	1+1+1
16	Dissolving and crystallising go on at equal rates (1); No net change (1); Hence amount of solute stays constant (1).	1+1+1
17	More evaporation of water (1); Salt remains and concentration increases (1); Dead Sea example (1).	1+1+1
	OR: Solution: smallest, no Tyndall (1); Colloid: intermediate, Tyndall (1); Suspension: largest, settles (1).	1+1+1
18	High pressure forces more gas into the liquid (1); Opening reduces pressure (1); Gas escapes (1).	1+1+1
	OR: They do not dissolve in water (1); Special solvents dissolve them (1); Like dissolves like (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Kitchen: tea, syrups (1); Medicine: ORS, antiseptics (1); Industry: dyes, cleaning (1); Agriculture: fertiliser solutions (1); Concentration is important (1).	1+1+1+1+1
	OR: Sugar, salt: soluble (1); Chalk: insoluble (1); Oil: insoluble in water (1); Alcohol: miscible with water (1); Like dissolves like (1).	1+1+1+1+1
20	Take equal masses of both (1); Same volume and temperature (1); Stir the same way (1); Measure time to dissolve (1); Powder dissolves faster (1).	1+1+1+1+1
	OR: Concentrated: more solute (1); Dilute: less solute (1); Mass percentage formula (1); Example: 20 g in 80 g water gives 20 per cent (1); Dilution reduces percentage (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.